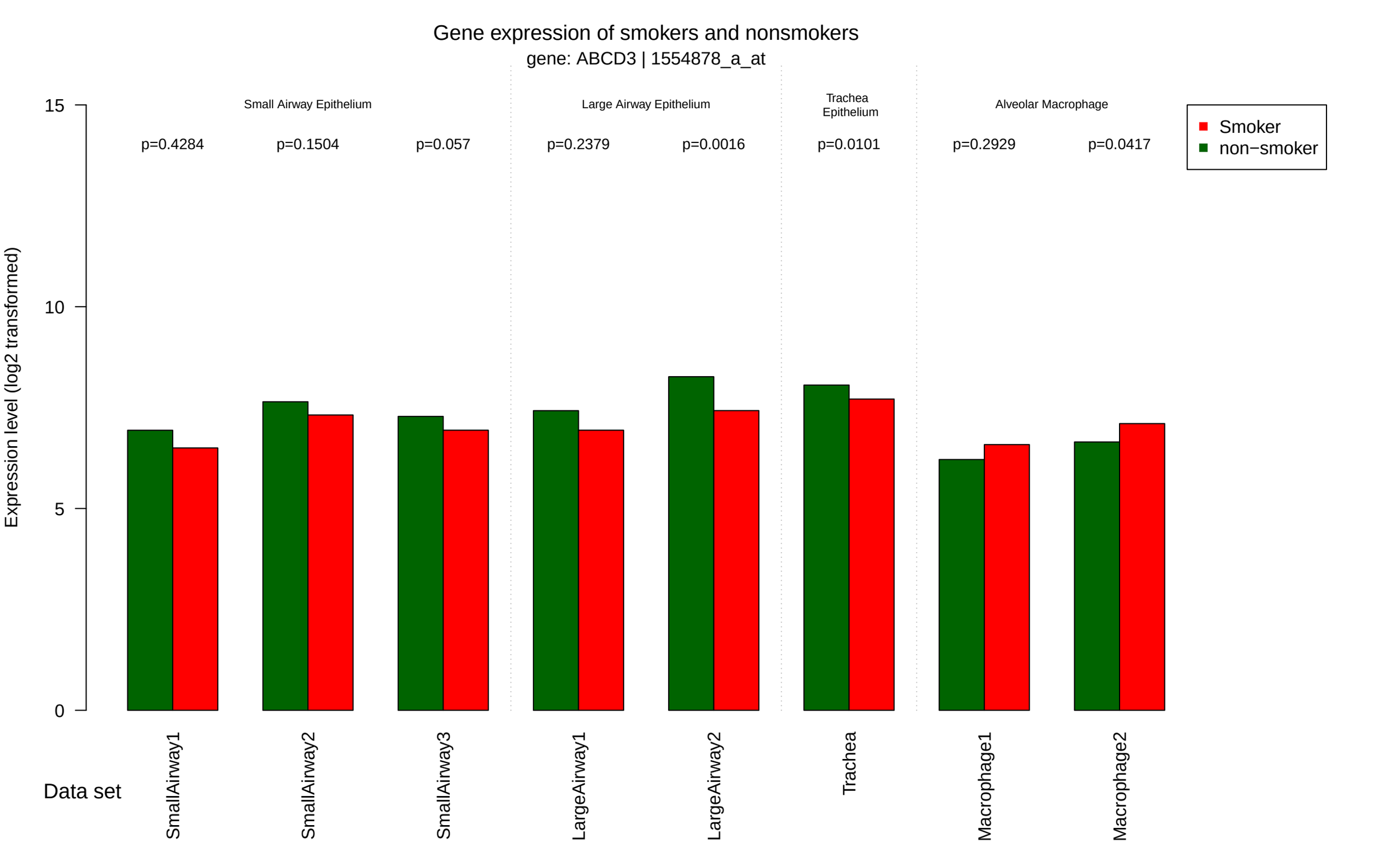
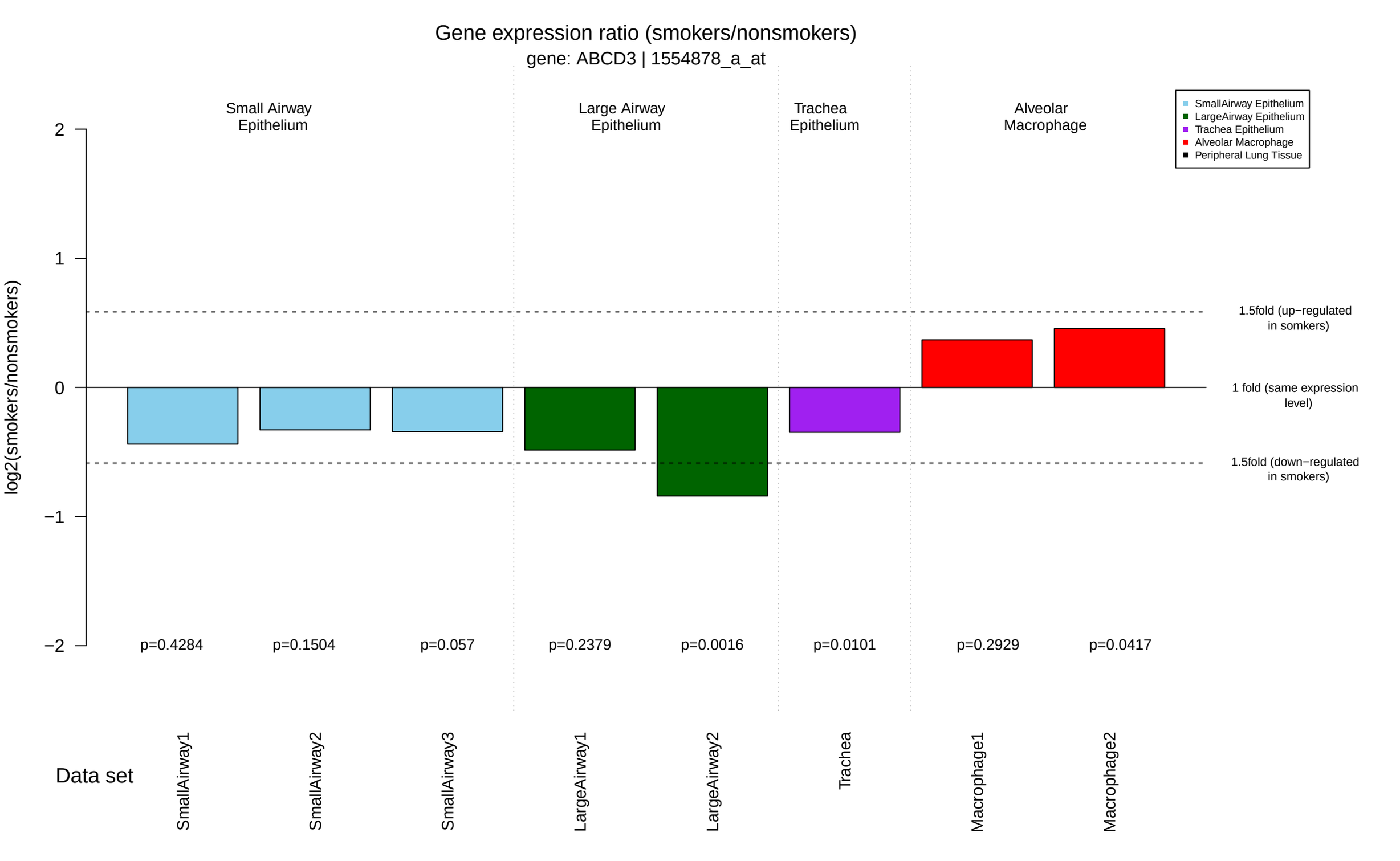
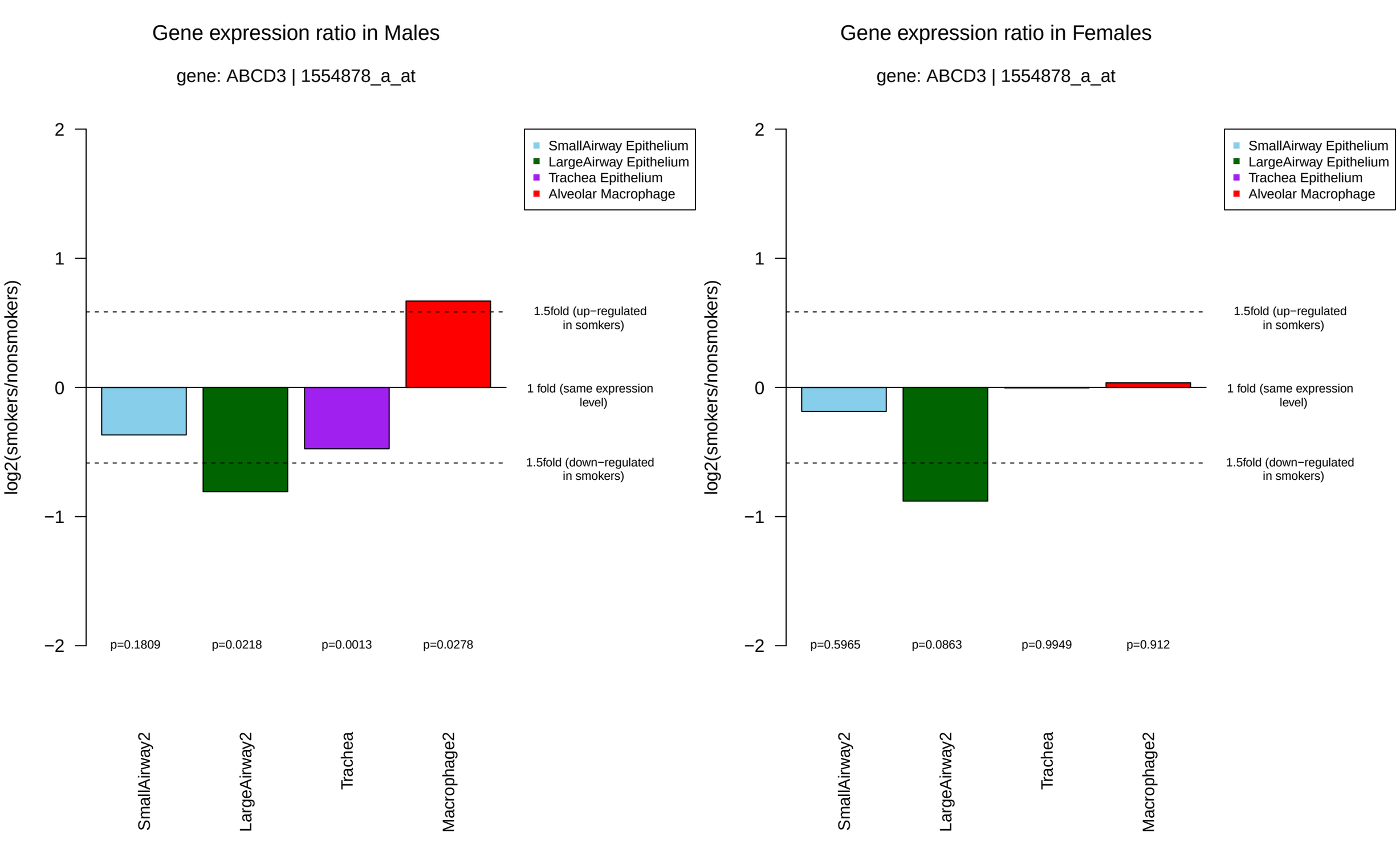
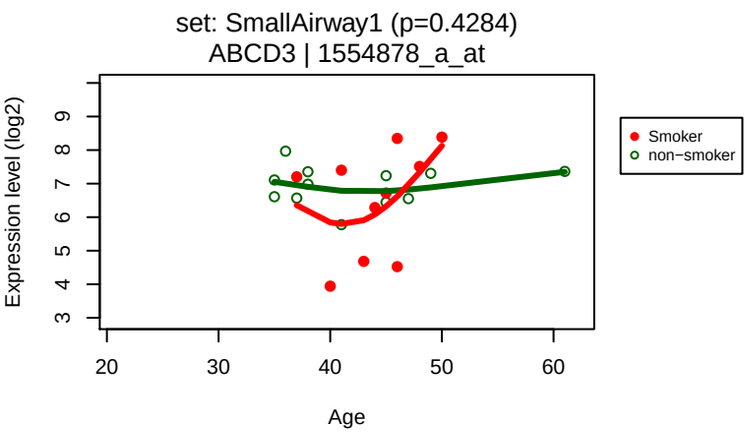
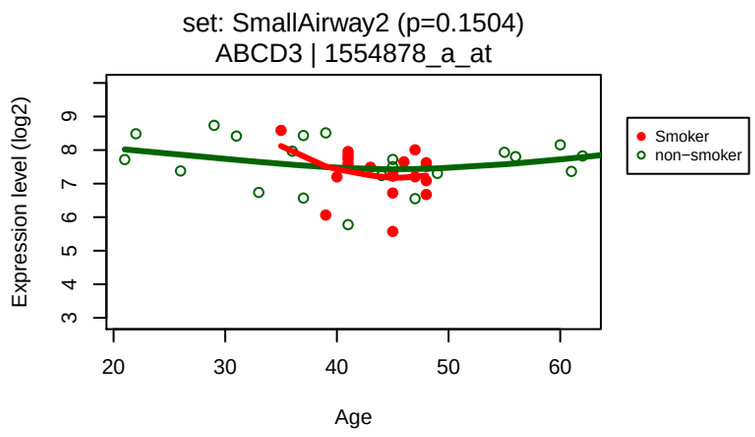
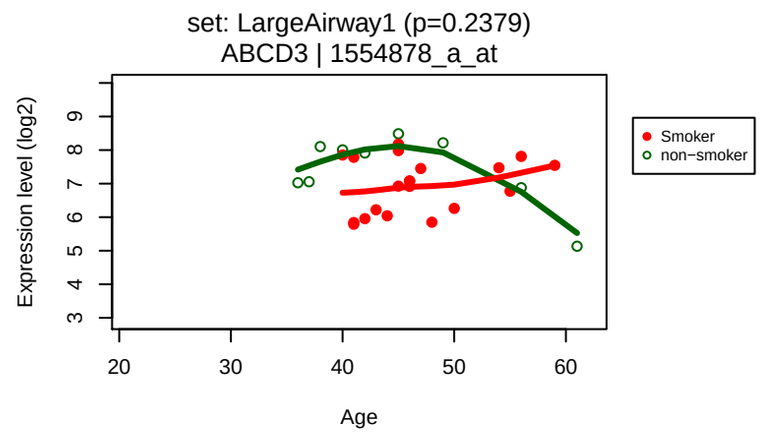

Gene expression of smokers and nonsmokers
gene: ABCD3 | 1554878_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4284
p=0.1504
p=0.057
p=0.2379
p=0.0016
p=0.0101
p=0.2929
p=0.0417
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

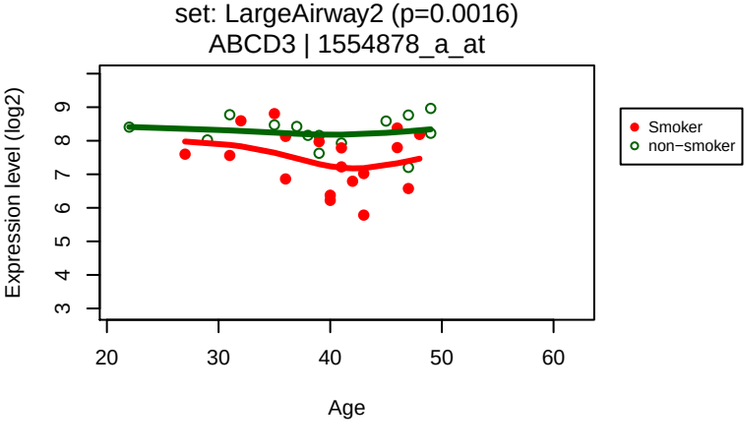

Gene expression ratio (smokers/nonsmokers)
gene: ABCD3 | 1554878_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4284
p=0.1504
p=0.057
p=0.2379
p=0.0016
p=0.0101
p=0.2929
p=0.0417
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

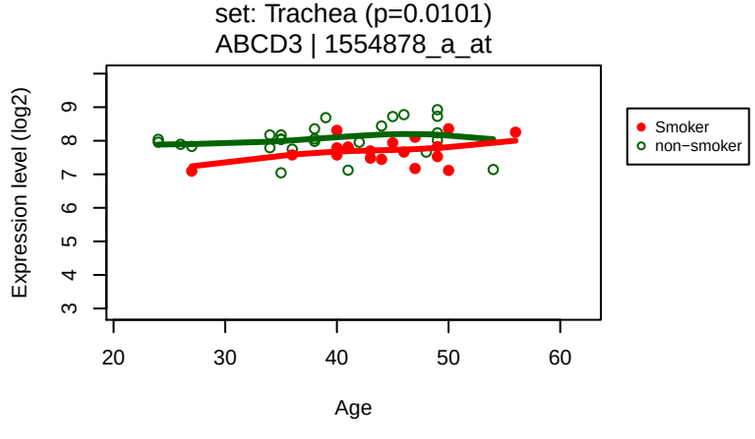

Gene expression ratio in Males
gene: ABCD3 | 1554878_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1809
p=0.0218
p=0.0013
p=0.0278
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ABCD3 | 1554878_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5965
p=0.0863
p=0.9949
p=0.912
SmallAirway2
LargeAirway2
Trachea
Macrophage2

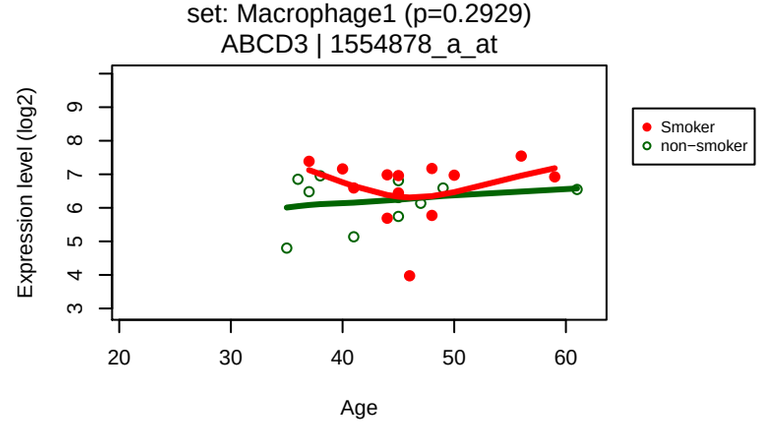

set: SmallAirway1 (p=0.4284)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker

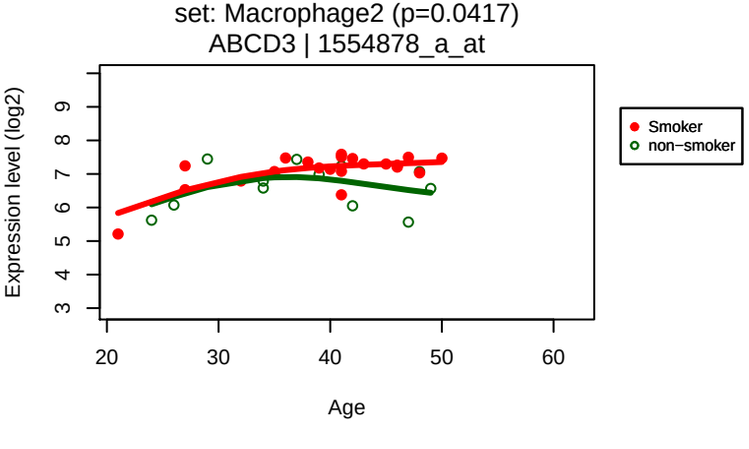

set: SmallAirway2 (p=0.1504)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker

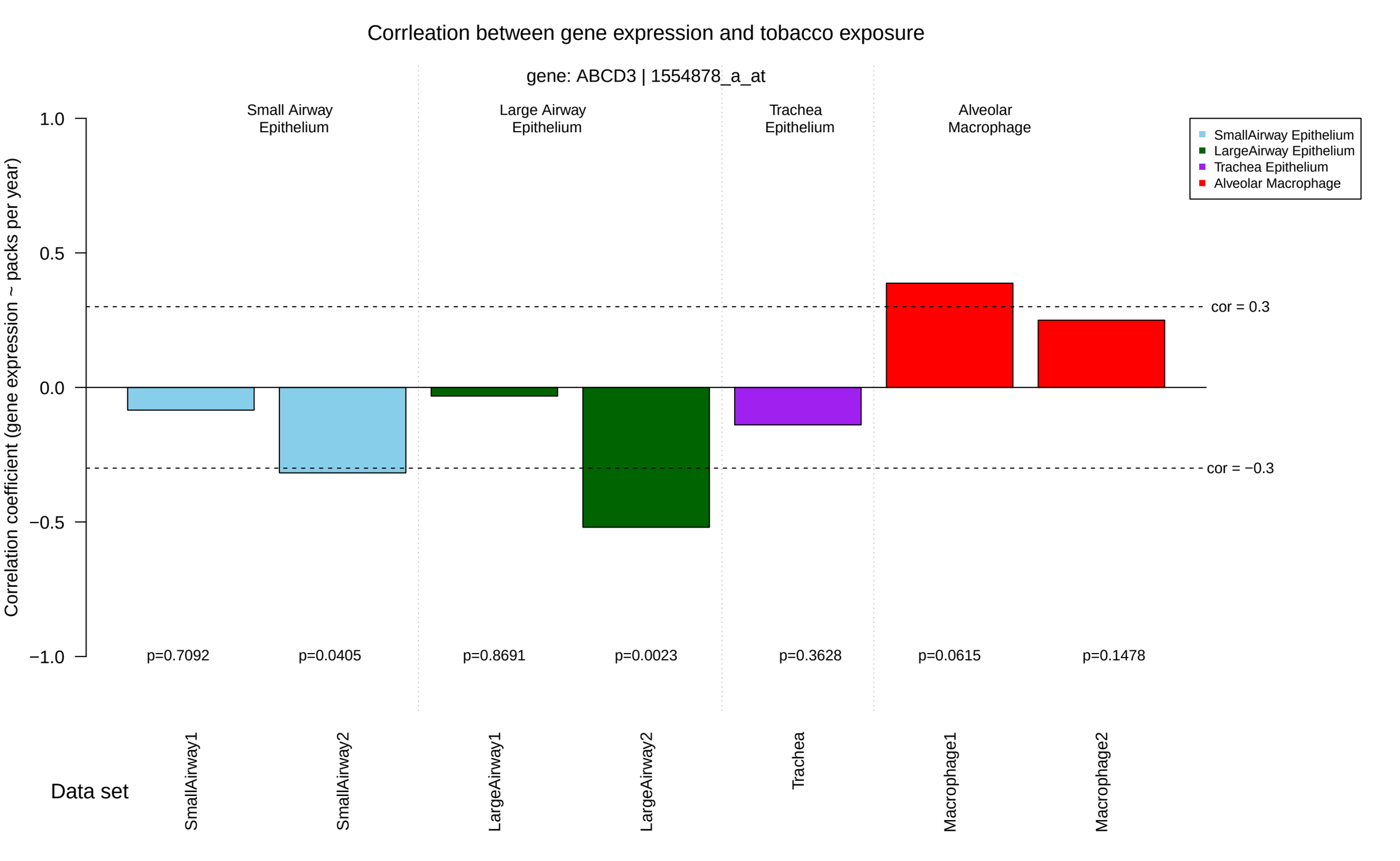

set: LargeAirway1 (p=0.2379)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0016)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0101)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2929)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0417)
ABCD3 | 1554878_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ABCD3 | 1554878_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.7092
p=0.0405
p=0.8691
p=0.0023
p=0.3628
p=0.0615
p=0.1478
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2